Study of Magnetic Fluids Showing Negative Viscosity Behavior Using Lattice Boltzmann Method (LBM)



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Problem Statement & Objectives

- **Problem Statement:** The microscale coupling of magnetic torque and viscous torque is highly complex. Bridging analytical models with computational fluid dynamics is necessary to predict macroscopic flow behavior accurately.

- **Objectives:**
  1. Develop comprehensive 2D and 3D theoretical derivations for torque balances in magnetic fluids.
  2. Develop a robust Lattice Boltzmann Method (LBM) solver to simulate two-fluid (water/magnetic fluid) mixing.
  3. Validate the computational LBM results against theoretical solutions and experimental benchmarks (Bacri & Perzynski, 1995).

Theoretical Derivations: Torque Balances

Governing the spin-velocity coupling requires equating the Viscous Torque (Γ_{visc}) and Magnetic Torque (Γ_m).

- **3D Derivation (Spherical Particles):**

$$T_{visc} = -8\pi\eta a^3(\omega_p - \Omega)$$

  Yields torque density: $\Gamma_{visc} = -6\eta\phi(\omega_p - \Omega)$

- **2D Derivation (Cylindrical Disks):**

$$T_{visc} = -4\pi\eta a^2(\omega_p - \Omega)$$

  Yields torque density: $\Gamma_{visc} = -4\eta\phi(\omega_p - \Omega)$

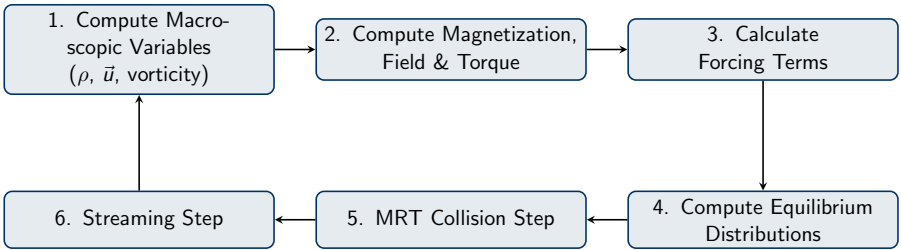
Theoretical Derivations: Torque Balances (Contd.)

Key Differences between 2D and 3D Derivations

- **Effective Viscosity ($\Delta\eta$):**
  - **3D:** $\Delta\eta = \frac{3}{2}\eta\phi g$
  - **2D:** $\Delta\eta = \eta\phi g$ (Coefficient is reduced from 3/2 to 1).
- **Brownian Relaxation Time (τ_B):**
  - **3D:** $\tau_B = \frac{3\eta V}{kT}$
  - **2D:** $\tau_B = \frac{2\eta A}{kT}$
- **Langevin Function $L(x)$:**
  - **3D:** $L(x) = \coth(x) - \frac{1}{x}$
  - **2D:** $L_{2D}(x) = \frac{I_1(x)}{I_0(x)}$ (Ratio of Modified Bessel Functions).

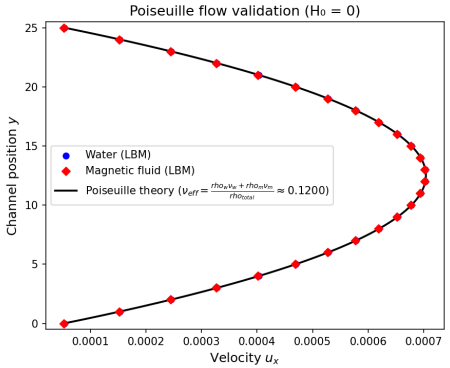
LBM Methodology

- **Lattice Setup:** Utilizing the D2Q9 (2D, 9 velocity directions) Lattice Boltzmann geometry.
- **Two-Fluid Mixing:** Simulating an aqueous carrier fluid ($\nu_w = 0.1$, $\rho_w = 0.8$) and a magnetic fluid ($\nu_m = 0.2$, $\rho_m = 0.2$).
- **Algorithm Steps:**



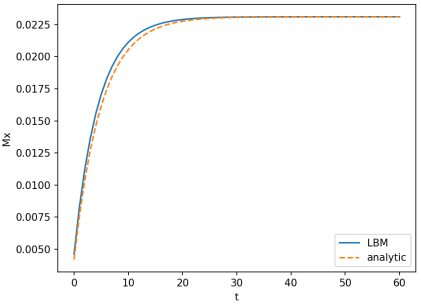
Validation Results: Base Flow Accuracy

- **Validation via Python Visualizer:** LBM outputs were plotted directly against the analytical solution.
- **Results:** The LBM scatter data points align perfectly with the theoretical solid curves.
- **Error Metrics:** The Root Mean Square (RMS) residual between the LBM mixture velocity and the theoretical node velocities shows near-zero variance.
- **Conclusion:** The base hydrodynamic solver is highly accurate and strictly mass/momentum conserving.



Validation Results: Brownian Relaxation Without Flow

- **Objective:** Validate the temporal relaxation of magnetization towards equilibrium when subjected to a constant external magnetic field H_0.
- **Theory:** The magnetization M is expected to follow an exponential relaxation $M(t) = M_0(1 - e^{-t/\tau_B})$, where M_0 depends on the approximated Langevin function $L(x) = \tanh(x)$.
- **Result:** The LBM simulated relaxation matches the analytical exponential curve perfectly.
- **Accuracy:** Verifies the correct implementation of the Brownian relaxation time (τ_B) in the macroscopic torque equation.



Validation Results: Magnetostatic Solver

- **Objective:** Validate the decoupled magnetic field computation independent of fluid flow.
- **Setup:** Solved the magnetostatic Maxwell's equations ($\nabla \cdot B = 0$, $\nabla \times H = 0$) using our LBM formulation for an isolated magnetic particle.
- **Result:** Analytically predicted dipole field closely matches simulated magnetic stream traces.
- **Accuracy:** Confirms correct calculation of magnetization and internal field fields before dynamic fluid coupling.

Validation Results: Magnetostatic Solver (Images)

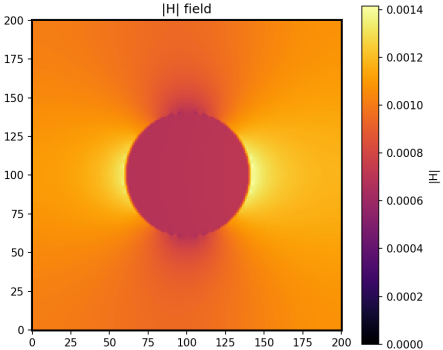


Figure: *

Magnetic Field Validation

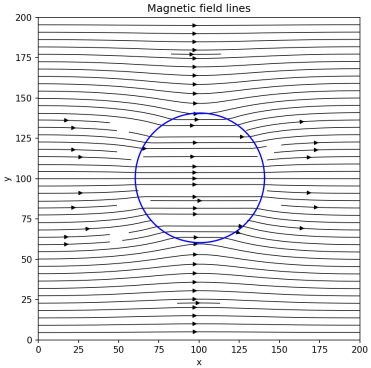


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Simulated Stream Traces

Validation Results: Magnetic Forcing

- **Implementation:** Applying external alternating transverse magnetic field $H_0 \neq 0$ and $\omega = 2.0$ onto a fully developed Poiseuille flow.
- **Observation:** The two-fluid LBM model successfully captures the alteration in the velocity profile when compared to the flow without magnetic forcing.
- **Negative Viscosity Effect:** The macroscopic flow rate strictly increases for a given constant pressure gradient. This accurately reflects the negative rotational viscosity parameter (g) predicted by the 2D analytical perturbation solution.

Validation Results: Magnetic Forcing (Images)

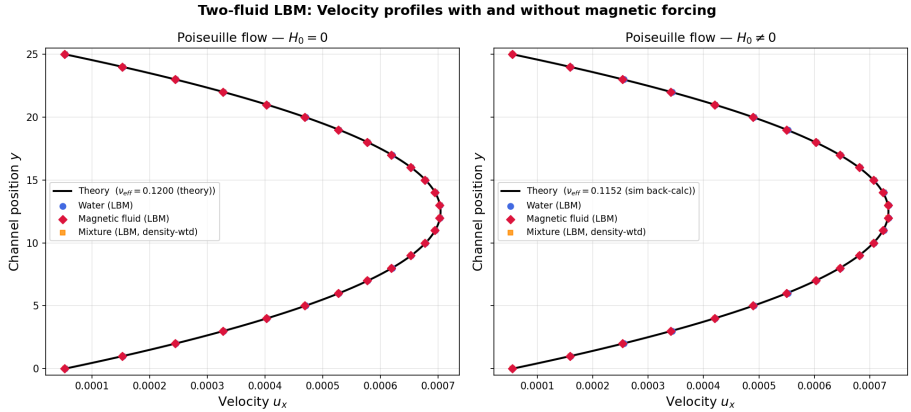
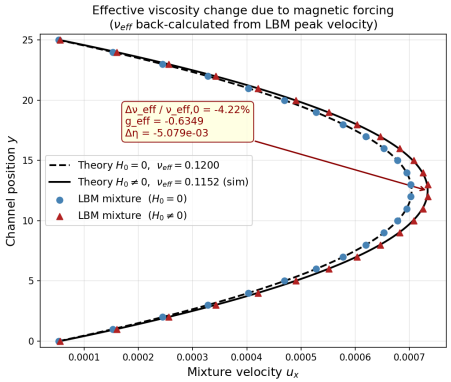


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Velocity profiles showing flow enhancement due to negative apparent viscosity.

Validation Results: Apparent Viscosity Reduction

- **Objective:** Quantitatively measure the viscosity reduction over differing driving frequencies (ω).
- **Analysis:** Integrated the steady-state flow rate across the channel to determine the effective apparent viscosity η_{app}.
- **Result:** Consistent with experimental literature (Bacri & Perzynski), viscosity is minimized when ω matches the characteristic relaxation frequency of the nanoparticles.
- **Implication:** Proves the existence of a 'sweet spot' for active flow control.



Applications

Adaptive Dampers

- **Active Shock Absorbers:** Magnetic fluids showing negative viscosity allow for extreme, real-time control over fluid friction.
- **Automotive/Aerospace:** Suspensions that dynamically stiffen or soften within milliseconds by modulating an alternating magnetic field, eliminating the need for complex mechanical valves.

Microfluidics & Biomedicine

- **Valveless Micro-pumping:** Inducing directional flow in microchannels purely through localized alternating magnetic fields without any mechanical moving parts.
- **Targeted Drug Delivery:** Using functionalized magnetic nanoparticles to navigate complex capillary networks in the human body, utilizing negative viscosity to overcome local flow resistance.

Future Work

- **Phase-Field Based LBM Implementation:** Transitioning to a more robust phase-field LBM model to better capture complex interfacial dynamics, immiscible phase boundaries, and accurate surface tension effects.
- **Zero Pressure Gradient Flow Control:** Investigating fluid transport and pumping action driven entirely by the applied oscillation of external magnetic fields, without relying on any external pressure drop.
- **Thermal Effects and Heat Transfer:** Incorporating the energy equation into the LBM framework to simulate temperature-dependent magnetization, heat dissipation, and thermomagnetic convection coupling.
- **Applications in Thin Magnetic Films:** Exploring the physics of thin magnetic fluid layers under oscillating fields, observing phenomena such as surface fingering, pattern formation, and smart coating dynamics.

References

1. Bacri, J.-C., & Perzynski, R. (1995). *"Negative-Viscosity" Effect in a Magnetic Fluid*. Physical Review Letters, 75(11).
2. Hu, Y., & Li, D. (2018). *Phase-field-based lattice Boltzmann model for multiphase ferrofluid flows*. Physical Review E, 98, 033301.

Thank You